TAMIL NADU STATE BOARD - STANDARD 9 SCIENCE (CHEMISTRY)

CHAPTER 3: STRUCTURE OF ATOM

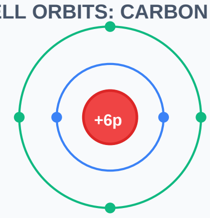
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 9 English Medium
Subject:	Science (Chemistry)
Chapter Name:	Structure of Atom

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

TOPIC ROADMAP: BOHR ATOMIC SHELL ORBITS: CARBON (Z=6)



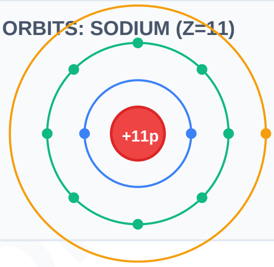
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Cathode Rays	Streams of fast-moving electrons emitted from the negative electrode (cathode) in a vacuum discharge tube.
Valency	The combining capacity of an element, determined by the number of electrons lost, gained, or shared to achieve a stable octet.
Isotopes	Atoms of the same element having the same atomic number (protons) but different mass numbers (neutrons).
Isobars	Atoms of different elements having different atomic numbers but the same mass number.
Isotones	Atoms of different chemical elements having different atomic numbers but sharing the same number of neutrons.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

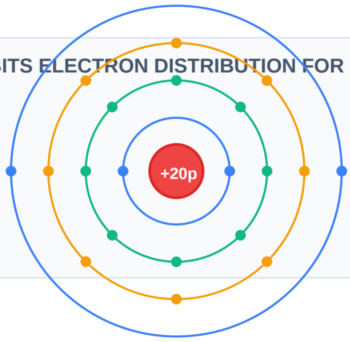
GLOSSARY: BOHR ATOMIC SHELL ORBITS: SODIUM (Z=11)



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

FORMULA: BOHR ATOMIC ORBITS ELECTRON DISTRIBUTION FOR CALCIUM (Z=20)



Essential Formulas, Laws & Identities:

- Bohr Orbit Electron Capacity = 2 * n^2 where n is the orbit shell number (K=1, L=2, M=3, N=4)
- Mass Number (A) = Number of Protons (Z) + Number of Neutrons (N)
- Number of Neutrons (N) = Mass Number (A) - Atomic Number (Z)

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: An element has atomic number Z = 11 and mass number A = 23. Find the number of protons, electrons, and neutrons.

Step-by-step Solution:

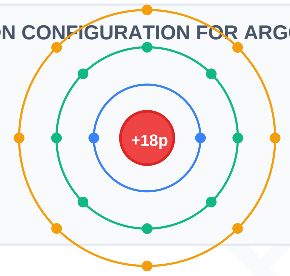
Atomic number Z = 11, so Protons = 11 and Electrons = 11. Neutrons = A - Z = 23 - 11 = 12.

Example 2: Write the electronic configuration of Calcium (Z = 20) using the Bohr-Bury rules.

Step-by-step Solution:

Z = 20. Shell capacities: K=2, L=8, M=8, N=2. Note that the outermost shell cannot have more than 8 electrons, so the 9th and 10th electrons go to N instead of completing M first. Configuration: 2, 8, 8, 2.

WORKED: BOHR SHELL ELECTRON CONFIGURATION FOR ARGON (Z=18)



HOTS (Higher Order Thinking Skills) Challenge

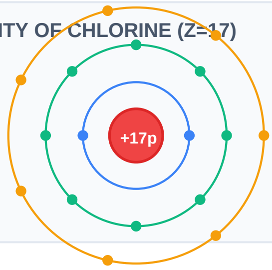
Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: An ion X^2- has 10 electrons and 8 neutrons. Identify the element and calculate the mass number of its neutral atom.

Analytical Solution Strategy:

The ion X^2- has 10 electrons, which means the neutral atom has 10 - 2 = 8 electrons. Therefore, its atomic number Z = 8 (Oxygen). Mass number A = Z + Neutrons = 8 + 8 = 16.

HOTS: VALENCE SHELLS REACTIVITY OF CHLORINE (Z=17)

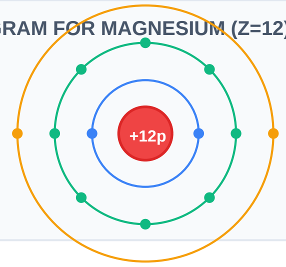


Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

- **Q1 (1-Mark):** Explain Rutherford's Alpha Particle Scattering Experiment and its key atomic model postulates.
- **Q2 (2-Mark):** State the main drawbacks of Rutherford's nuclear model and how Bohr's model resolved them.
- **Q3 (2-Mark):** Draw the electronic Bohr shell diagrams for: Carbon, Sodium, and Chlorine.
- **Q4 (5-Mark):** Compare isotopes, isobars, and isotones with two examples of each.
- **Q5 (5-Mark):** Explain the applications of isotopes in medicine (cancer treatment, thyroid scanning).

PRACTICE: VALENCE SHELL DIAGRAM FOR MAGNESIUM (Z=12)



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Cancer Therapy

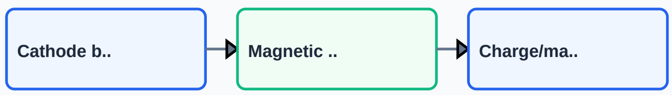
Cobalt-60 isotope radiation is targeted at cancer cells to destroy malignant tissue without major surgical operations.

Application Case: Archaeological Dating

Carbon-14 isotope half-life measurements are used to determine the age of ancient fossils and wooden carvings.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

APPLICATION: THOMSON'S CRT TUBE ELECTRON DISCOVERY



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.

✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.

✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3, N) for the final numeric answer.

✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

[ ] Read and memorized core definitions on Page 2.

[ ] Reviewed critical formulas and relations on Page 3.

[ ] Solved worked examples on Page 4.

[ ] Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress. Good luck!

REVISION: CHEMISTRY ATOMIC ORBITS ELECTRON CONFIGURATION